

PRODUCT OVERVIEW

SevaSaathi

Verified Home Care & Elder Support Platform

Technical Architecture & Platform Capabilities

VERSION 1.0

2025

SECTION 01

The Problem

India is undergoing a profound demographic transformation. By 2026, the country will be home to over **140 million elderly citizens**, making it one of the fastest-aging populations in the developing world. Yet the infrastructure to support this shift remains dangerously inadequate. Unlike Western nations that built elder care systems over decades, India faces this challenge with fragmented services, no standardized quality benchmarks, and a workforce that is largely unorganized and unverified.

Families seeking home care for aging parents or recovering patients confront a landscape of uncertainty. There is **no trusted marketplace** to find qualified caregivers. Most families rely on word-of-mouth referrals from neighbors or local agencies with zero accountability. When a caregiver underperforms, neglects medication schedules, or mistreats an elderly parent, there is no recourse, no review system, and no oversight body to turn to. The emotional toll on families already dealing with the stress of caregiving is immense.

Pricing is equally opaque. Caregiver fees vary wildly between providers with no correlation to quality or experience. Families cannot compare options, verify credentials, or track the care their loved ones receive in real time. The entire ecosystem operates on **trust without verification**, and the consequences fall on the most vulnerable members of society. SevaSaathi exists to solve every one of these pain points with a single, comprehensive digital platform.

SECTION 02

The Solution: SevaSaathi

SevaSaathi is a **complete digital platform** that connects families in need of home care with thoroughly verified, qualified caregivers. Unlike generic classifieds or unstructured referral networks, SevaSaathi provides an end-to-end experience: from creating a detailed patient profile with medical history and dietary requirements, through AI-powered smart matching that considers skill sets, location proximity, experience levels, and real-time availability, all the way to secure payment processing and post-care reviews.

At its core, the platform features a **real-time care monitoring system** powered by Socket.io. Caregivers submit daily care reports documenting the patient's activities,

mood, meals, and medication adherence. Families receive instant notifications and can monitor care quality as it happens, not days later. This transparency layer transforms home care from a blind trust exercise into a data-driven, accountable service delivery model.

The platform is built on a modern tech stack featuring Next.js 16 with React 19, TypeScript, and Tailwind CSS 4. It runs as a Progressive Web App for offline access, supports multiple authentication methods including Google OAuth and phone OTP, and integrates AI-powered document verification using the z-ai-web-dev-sdk for Aadhaar OCR processing. Every design decision prioritizes reliability, speed, and the specific needs of the Indian home care market.

SECTION 03

Key Platform Features

AI ENGINE

Smart Matching Engine

A 5-dimension weighted algorithm evaluates caregivers across skill match (30%), location proximity (25%), experience level (20%), real-time availability (15%), and historical rating (10%). This produces a composite relevance score that surfaces the best-fit caregivers first, eliminating guesswork for families.

DASHBOARDS

Multi-Role Interface

Three fully tailored dashboards serve Families, Caregivers, and Administrators. Each role sees only the features and data relevant to their workflow. Families manage patients and bookings, caregivers handle schedules and reports, admins oversee the entire platform with analytics and approval tools.

REAL-TIME

Live Care Monitoring

Socket.io powers instant bidirectional communication between caregivers and families. When a caregiver submits a daily care report, families receive push notifications immediately. The system tracks activities, mood changes, food intake, and medication compliance in structured, reviewable reports.

PAYMENTS

Secure Payment System

UPI deep link integration enables instant payments via any Indian UPI app. Razorpay is integrated as a secondary gateway for maximum coverage. The platform charges a transparent 10% fee, with 90% going directly to the caregiver. Full payment history and withdrawal tracking are built in.

VERIFICATION

AI Document Verification

Caregivers must verify their identity through 7 document types including Aadhaar, PAN, and professional certificates. AI-powered Aadhaar OCR extracts and validates data automatically using the z-ai-web-dev-sdk.

Administrators review and approve or reject submissions before a caregiver can receive bookings.

RESOLUTION

Complaint & Resolution

A structured complaint system with priority levels (Low, Medium, High, Critical) ensures every issue is tracked to resolution. Administrators are auto-assigned based on workload, and families can escalate unresolved complaints. The entire lifecycle is logged for transparency and accountability.

SECTION 04

How It Works

The SevaSaathi workflow is designed to be intuitive for families while maintaining rigorous quality controls on the caregiver side. Here is the complete journey from search to sustained care:

1

Create Patient Profile

The family registers and creates a detailed patient profile including medical history, dietary requirements and restrictions, mobility status, and any special care needs. This data powers the AI matching engine and ensures caregivers arrive fully informed.

2

AI Matches Best Caregivers

The smart matching algorithm scores every available caregiver across five weighted dimensions. Results appear ranked by composite fit score, with detailed breakdowns so families understand why each match was suggested and can make informed decisions.

3

Select & Book Care

Families review matched caregiver profiles, read ratings from previous clients, and select their preferred caregiver. Booking supports multiple shift types including day shifts, night shifts, 12-hour, and 24-hour care arrangements with flexible scheduling.

4

Daily Care Reports

The confirmed caregiver provides structured daily reports covering activities performed, patient mood and behavior, meals consumed, and medications administered. These reports are timestamped and create a comprehensive care history accessible to the family at any time.

5

Monitor & Review

Families monitor care quality in real-time through the dashboard. After a care period concludes, they submit multi-dimensional reviews covering punctuality, care quality, communication, and overall satisfaction. A complaint filing system is available for any concerns.

6

Secure Payment & Payout

Payments flow through the platform with full transparency. Families pay via UPI or Razorpay, the platform retains 10% as a service fee, and the remaining 90% is available for the caregiver to withdraw. Complete transaction histories are maintained for both parties.

Matching Algorithm Breakdown



SECTION 05

Technology Stack

SevaSaathi is built on a carefully selected modern technology stack optimized for developer productivity, application performance, and the specific requirements of the Indian market. Every technology choice is deliberate and purpose-driven.

FRONTEND

Next.js 16 with React 19 and TypeScript 5. UI built with Tailwind CSS 4 and shadcn/ui component library. Animations via Framer Motion. Progressive Web App enabled for offline access.

BACKEND

Next.js API Routes for server-side logic. Prisma ORM for type-safe database access. SQLite for development, MySQL for production. Full REST API with 30+ endpoints.

AI & MACHINE LEARNING

z-ai-web-dev-sdk powers Aadhaar OCR extraction and Vision AI capabilities. Automated document data parsing reduces manual verification effort and accelerates caregiver onboarding.

REAL-TIME

Socket.io standalone service provides bidirectional live communication. Instant care report delivery, notification push, and real-time booking status updates to all connected clients.

COMMUNICATION

Gmail SMTP via Nodemailer for transactional emails. Fast2SMS API for Indian phone number SMS delivery. Email logging for audit trails and delivery tracking.

AUTHENTICATION

NextAuth.js v4 with JWT session management. Google OAuth, email OTP, and phone OTP login methods. bcrypt password hashing with 10 salt rounds for security.

INFRASTRUCTURE

Progressive Web App with service worker caching. Caddy reverse proxy for production deployment. Standalone Node.js server with horizontal scaling capability.

PAYMENTS

UPI deep link integration for native app payments. Razorpay gateway as secondary payment processor. Automated fee splitting with 10% platform commission.

SECTION 06

User Roles & Capabilities

SevaSaathi serves three distinct user types, each with a purpose-built dashboard and feature set tailored to their specific needs and workflows in the home care ecosystem.

For Families

Families gain complete control over the care journey. They create and manage detailed **patient profiles** including medical history, dietary requirements, and mobility status. The **smart caregiver search** surfaces AI-matched professionals ranked by fit score. **Booking management** covers creating, tracking, cancelling, and requesting urgent care with flexible shift types. **Real-time care reports** deliver daily updates on activities, mood, meals, and medications. After each engagement, families submit **multi-dimensional reviews** and can file structured complaints through the resolution system.

For Caregivers

Caregivers build and manage a **professional profile** showcasing skills, certifications, availability, and a personal bio. The **weekly schedule system** allows setting day or night shift availability per weekday for granular control. Before accepting bookings, caregivers complete **document verification** covering Aadhaar, PAN, and professional certificates with AI-assisted processing. During active care, they submit detailed **care reports** tracking all patient activities. An **earnings dashboard** provides full financial transparency with withdrawal capabilities.

For Administrators

Administrators operate the platform from a powerful **analytics dashboard** showing user growth, booking trends, and revenue metrics. **User management** includes cascade operations for bulk actions across families, caregivers, and patients. The **verification workflow** lets admins review and approve or reject document submissions. **Complaint resolution** tools provide priority-based tracking with workload-balanced assignment. **Withdrawal processing** enables approval and payout of caregiver earnings with full audit trails.

SECTION 07

Security & Trust

Trust is the foundation of any care platform. SevaSaathi implements multiple layers of security and verification to ensure that only qualified, authenticated caregivers interact with vulnerable patients, and that all data remains protected at every stage.

- **Multi-Layer Authentication:** Users can sign in via Google OAuth for convenience, email OTP for security, or phone OTP for accessibility in India where email adoption varies. Each method is fully integrated with NextAuth.js v4 session management.
- **Password Security:** All passwords are hashed using bcrypt with 10 salt rounds, providing robust protection against brute-force attacks even in the event of database compromise.
- **JWT Session Management:** Sessions use JSON Web Tokens with a 30-day expiry. Tokens are validated on every API request, ensuring that expired or tampered sessions are immediately rejected.
- **Document Verification Gate:** Caregivers cannot receive any bookings until their documents are submitted, AI-processed, and manually approved by an administrator. This three-stage verification prevents unverified individuals from entering the care pipeline.

- **Complaint Resolution System:** All complaints are logged with timestamps, priority levels, and assigned administrators. Families can track resolution progress, and the platform retains complete records for accountability.
- **Email Audit Trail:** Every transactional email sent through the platform is logged in a dedicated EmailLog model, creating a comprehensive communication audit trail for dispute resolution and compliance.

SECTION 08

Platform Statistics

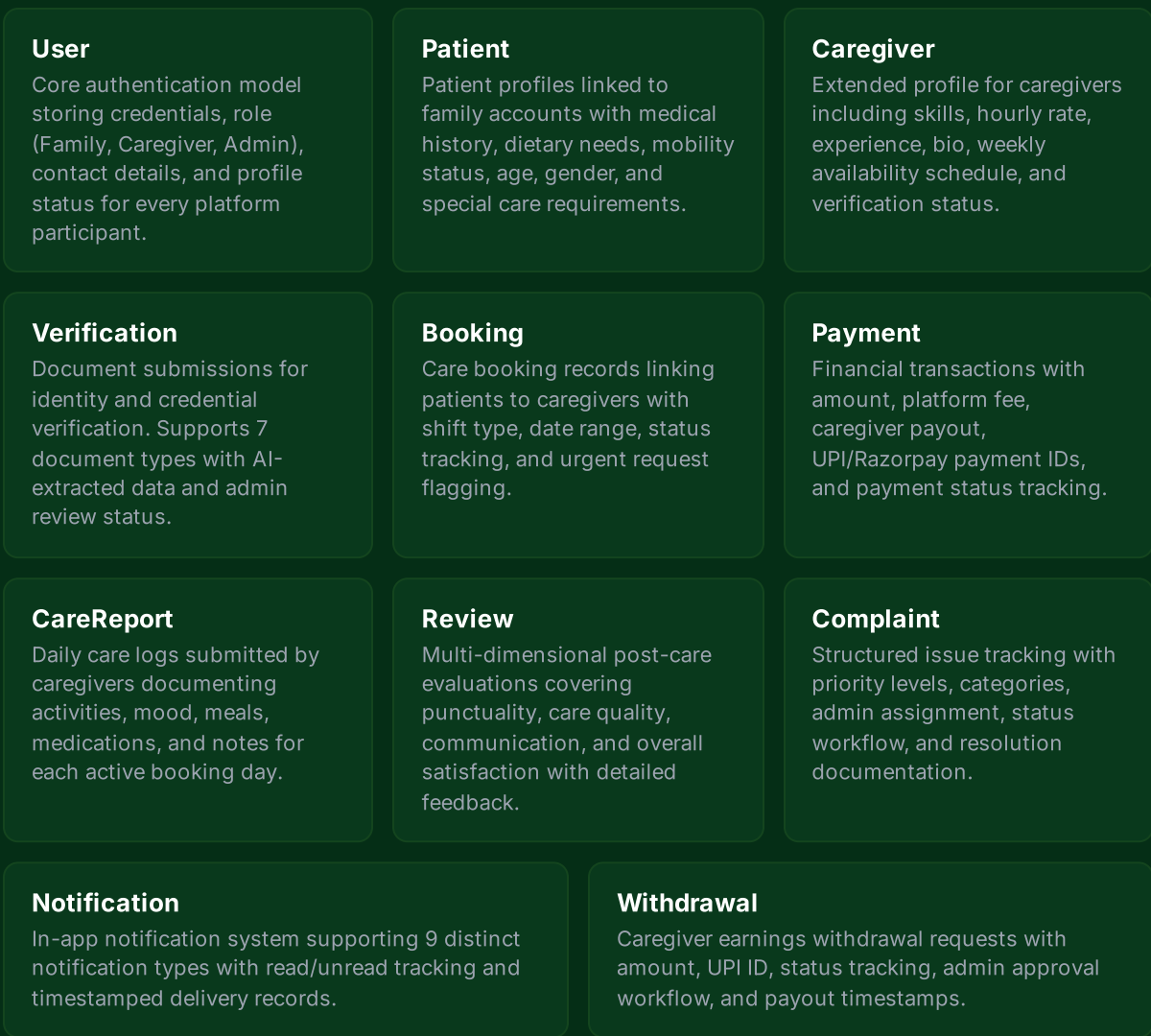
SevaSaathi is a substantial, production-grade platform built with meticulous attention to both breadth and depth. These numbers reflect the scale and completeness of the implementation.



SECTION 09

Database Architecture

The platform's data layer consists of 11 carefully designed Prisma models that cover every aspect of the home care workflow, from user registration and caregiver verification through bookings, payments, care delivery, and dispute resolution.



SECTION 10

Roadmap & Future Scope

SevaSaathi's current implementation represents a robust foundation. The following roadmap items are planned to expand platform capabilities, deepen market penetration, and unlock new revenue streams in the rapidly growing Indian home care market.

- **Public Ratings & Reviews:** Make caregiver ratings and reviews publicly visible on their profiles to build trust, enable informed decision-making, and incentivize high-quality care delivery through reputation systems.
- **Enhanced Admin Analytics:** Expand the administrator dashboard with advanced visualizations including booking trend charts, revenue analytics, geographic heatmaps of demand, and caregiver performance benchmarking tools.

- **Caregiver PWA:** Launch a dedicated mobile-first Progressive Web App for caregivers optimized for on-the-go report submission, schedule management, and instant notification receipt without requiring a desktop browser.
- **Automated Payout Processing:** Implement scheduled automatic payouts to caregivers based on completed bookings, reducing manual withdrawal processing overhead and improving caregiver satisfaction with faster payments.
- **Subscription Tiers:** Introduce Basic, Premium, and Enterprise subscription plans offering varying levels of features, priority matching, dedicated support, and advanced analytics for institutional clients.
- **Multi-Language Support:** Add support for Hindi, Tamil, Telugu, Bengali, and other major Indian languages to serve families and caregivers across diverse linguistic regions of the country.
- **Video Calling Integration:** Enable remote video consultations between families, caregivers, and healthcare professionals for initial assessments, follow-up check-ins, and emergency situations requiring visual evaluation.

Each roadmap item has been prioritized based on market demand, technical feasibility, and revenue impact. The platform's modular architecture ensures these features can be delivered incrementally without disrupting existing functionality, allowing SevaSaathi to grow alongside the evolving needs of India's home care ecosystem.

Let's Build the Future of Home Care Together

SevaSaathi is ready to transform how India cares for its elderly and recovering loved ones. We are actively seeking partners, investors, and early adopters to join this mission.

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SevaSaathi